

**EARLY PREDICTION OF CONSTRUCTION COST OVERRUNS USING
EXPLAINABLE MACHINE LEARNING ON PRE-CONSTRUCTION DATA WITH
EXTERNAL RISK INDICATORS**

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LIST OF ABBREVIATIONS

AI.....	Artificial Intelligence
ANN.....	Artificial Neural Network
AUC-ROC.....	Area Under the Receiver Operating Characteristic Curve
CPI.....	Consumer Price Index
DNN.....	Deep Neural Network
ETR.....	Extra Trees Regressor
GB.....	Gradient Boosting
GRU.....	Gated Recurrent Unit
LightGBM.....	Light Gradient Boosting Machine
LR.....	Linear Regression
LSTM.....	Long Short-Term Memory
ML.....	Machine Learning
NGBoost.....	Natural Gradient Boosting
PPI.....	Producer Price Index
R^2	Coefficient of Determination
RAM.....	Random Access Memory
RF.....	Random Forest
SHAP.....	SHapley Additive Explanations
SVM.....	Support Vector Machine
SVR.....	Support Vector Regression
VIF.....	Variance Inflation Factor
XGBoost.....	Extreme Gradient Boosting

Abstract

Construction cost overruns is one of the important challenges that affects the management, financial losses, schedule delays and project performance. The industry has some traditional cost estimation techniques which struggle to address this issue because those methods often rely on historical data and mostly will not consider the market conditions such as inflation, material price and labor cost, which keeps on changing. Those methods also struggle to capture the complex and nonlinear relationships between project characteristics and budget risks, especially before the execution phase of a project. Many studies applied machine learning techniques to predict construction cost overruns using the execution-phase data that will be available only during construction, which limits the early-stage decision-making process.

This research addresses this gap by developing an explainable machine learning classification framework to predict construction cost overruns using the early-stage data integrating with external economic indicators. This study will use two datasets, one containing the project level information such as project type, budget, location, resource allocation and duration. The second dataset contains the external economic time-series indicators and market related factors.

The framework will be developed by training the model with five supervised classification algorithms such as Logistic Regression, Random Forest, XGBoost, LightBGM and CatBoost to predict the cost overrun and then categorizing the projects into low, medium and high-risk categories for easier interpretation of the stakeholders. SHAP analysis will be applied on the models to identify the key cost drives which influence the cost overrun predictions to improve transparency and support in decision-making.

The expected output of this research will be an interpretable and decision-support framework that helps the project managers to identify the high-risk projects during the early-stage of the construction project. By incorporating economic indicators along with the early-stage data, this research helps the project managers to improve their planning before entering into execution phase and this helps in cost management.

1. Background

Construction cost overruns are a common issue in the industry, which affects the project management and can lead to significant financial losses and delays. Many construction projects exceed their estimated budgets due to inaccurate early estimates, design changes and unforeseen risks. Research shows that many construction projects experience some level of cost overrun, (Hamdan et al., 2025). This shows that traditional planning and estimation methods have struggled to address the cost overruns. So, improving cost prediction at the early stage of a project has become an important area of research.

In recent years, machine learning (ML) has been widely used to improve the accuracy of construction cost forecasting. These models can analyse complex patterns in large datasets and capture nonlinear relationships between project characteristics and final cost of the projects. Many existing studies depend on the data collected during the execution phase of projects. Hamdan et al., 2025 showed that execution-phase features can improve cost overrun prediction accuracy. Similarly, Al Mnaseer et al., 2025 used integrated approaches for project cost overruns and Khodabakhshian et al., 2024 achieved high accuracy but did not use the early-stage data. This limits the use of such models for early decision-making.

Some studies have shown that using ANN (Artificial Neural Networks), random forests, and gradient boosting techniques with early-stage project features can significantly improve prediction accuracy (Padala & Goyal, 2025; Hamdan et al., 2025). Similarly, (Deepa et al., 2025) studies proposed hybrid computing model combining multiple techniques like Artificial Neural Networks with Genetic algorithms that achieved high prediction accuracy on the cost estimation. These studies use pre-construction features, but they have not used any external economic factors which can influence the budget.

Some research included external factors into the models for cost prediction. For example, (Darko et al., 2023) incorporated environmental variables such as weather conditions into the prediction models. In addition, (Alzara et al., 2025) explored the impact of macroeconomic indicators such as inflation rates, material price indices, and labor market trends on

construction. These studies focused on time-series cost trends instead of early-stage predictions. Beyond accuracy and external factors, there is a concern about the interpretability of machine learning models in construction management. Studies such as (Gunarani et al., 2026; Chen et al., 2026; Das et al., 2024), apply explainability techniques like SHAP (Shapely Additive exPlanations), to identify key cost drivers which helped in more informed decision-making. While these methods improve interpretability, they often rely on subjective or execution-phase data, which affects their generalizability.

Finally, we can say that existing research shows that machine learning can improve cost prediction accuracy, but there is still a lack of models that combine early-stage data, external economic factors, and explainable AI techniques. This highlights the need for early-stage prediction framework that provides both accurate and interpretable insights managing cost overrun risks before execution phase.

2. Literature Review

The existing research on construction cost overrun prediction using machine learning techniques focuses on early-stage prediction models, the use of external factors and the application of explainable AI methods. By reviewing the existing research, the challenges and limitations can be identified which helps to find the research gap.

2.1 Early-Stage Cost Prediction Models Using Machine Learning

Several studies have explored the use of machine learning models in cost overrun predictions to improve accuracy in prediction. ANNs are the commonly used techniques in predicting cost. Padala & Goyal (2025) demonstrated ANNs with extensive hyperparameter tuning to predict early-stage costs and achieved lower prediction errors compared to traditional methods. Their approach demonstrated that ANNs are effective when applied to structured datasets with clearly defined input features such as project size, location, and material specifications. Similarly, Deepa et al. (2025) proposed hybrid models combining neural networks with optimization and uncertainty-handling methods to capture variability in construction cost data. However, these models improved accuracy by using internal project features and not used external economic factors which limits the ability to capture real-world cost variations.

2.2 High Accuracy Models Using Execution-Phase Data

Some studies achieved high accuracy by using execution-phase data. Hamdan et al. (2025) compared fifteen different machine learning algorithms and found that CatBoost achieved the highest explanatory power with an R^2 value exceeding 0.88. Their study highlighted the importance of feature selection which significantly improves the prediction performance when variables like variation orders are included. Similarly, Al Mnaseer et al. (2025) developed models for predicting both time and cost overruns by integrating metaheuristic optimization techniques with ANNs which shows that optimization algorithms can enhance model performance by efficiently navigating complex parameter spaces. However, these approaches depend on execution-phase data, which will not support for early decision-making.

2.3 Advanced Machine Learning and Uncertainty Modelling

Recent research has introduced more advanced machine learning frameworks that improve prediction. Chen et al. (2026) proposed a hybrid probabilistic learning model combining NGBoost and Extra Trees Regressor, achieving an R^2 of 0.9866 on a dataset of over 4,000 construction cost samples, and proving uncertainty estimation and SHAP-based interpretability. Similarly, Das et al. (2024) applied probabilistic boosting techniques with explainability to enhance prediction performance. While these studies used advanced techniques, they focuses on improving prediction accuracy and uncertainty quantification and is not addressing early-stage decision making using pre-construction data.

2.4 External Factors in Cost Prediction

Some studies used the external factors such as macroeconomic conditions, material price inflations, labor market trends in construction cost prediction. Darko et al. (2023) included environmental variables such as weather conditions, which was a good initiative however macroeconomic indicators like inflations, market trends were not included. Alzara et al. (2025) took this to a next step by including macroeconomic factors such as CPI, PPI, oil prices, foreign exchange rates and stock market indices while building the models. However, these approaches focused on cost trends or specific scenarios and did not integrate external factors with the pre-construction data.

2.5 Explainability and Interpretability in Cost Prediction

The interpretability of machine learning models has become an important area in the construction industry to gain the trust of the stakeholders. Gunarani et al. (2026) applied SHAP to identify the key cost drivers which increased transparency and provided more informed decision-making to the stakeholders. Similarly, Chen et al. (2026) considered explainability as an important requirement along with accuracy and uncertainty quantification by integrating SHAP analysis into their model. Similarly, Das et al. (2024) applied SHAP to NGBoost model for construction cost estimation which proved that the results support informed decision-making by showing which project features influence the costs. Even after these advancements, these models are often applied to mixed-stage or subjective datasets which did not use the pre-construction data in cost prediction.

2.6 Summary of Key studies

Table 1 summarises the reviewed studies by explaining their datasets, methodologies, and key findings and limitations.

Table 1: Survey of Reviewed Literature

S.No	Paper Title & Year	Dataset	Methodology	Key Findings	Limitations
1.	Padala & Goyal (2025) Early stage cost prediction model for Indian building construction projects using artificial neural networks	377 building projects(India), 17 early-stage features	Artificial Neural Networks(ANN) with hyperparameter tuning	High accuracy using Pre-Construction data	No external economic factors and limited interpretability
2.	Hamdan et al. (2025)- Predicting cost overrun in construction projects using machine learning algorithms: the case of Jordan	836 public projects(Jordan), 12 features including initial budget and execution-phase variables	CatBoost, XGBoost, Random Forest, Neural Networks(compared 15 algorithms)	Very high accuracy in cost overrun prediction	Relies on execution-phase data
3.	Deepa et al. (2025)- A hybrid machine learning approach for early cost estimation of pile foundations	180 pile foundation samples (India), design and site parameters	Hybrid computing model using ANN, Genetic Algorithms and Fuzzy Logic	High prediction accuracy on early cost estimation for pile foundations	Narrow scope, limited generalizability and lacks interpretability
4.	Darko et al. (2023)- Using machine learning to improve Cost and duration prediction accuracy	198 green building projects (Hong Kong), planning phase features	Deep Neural Networks (DNN) and Support Vector Regression (SVR)	Improved cost prediction with environmental factors	No macroeconomic indicators

5.	Alzara et al. (2025)- Macroeconomic-aware forecasting of construction costs in developing countries: Using gated recurrent unit and long short-term memory deep learning framework	National construction cost index data (Egypt), macroeconomic indicators	LSTM, GRU- deep learning for time-series forecasting	Strong impact of economic indicators on costs	Focus on time-series, not project-level prediction
6.	Gunarani et al. (2026) – A Machine Learning-Driven Framework for Sector-Specific Construction Cost Overrun Prediction and Mitigation	70 survey responses, 15 perceptual cost overrun factors	Random Forest, XGBoost, CatBoost and SHAP for explainability	Sector-specific cost overrun prediction and identifies key cost drivers using explainability	Used subject or mixed stage data, not early-stage data
7.	Chen et al. (2026)- Uncertainty aware and explainable construction cost prediction using a hybrid probabilistic learning model	RSMeans dataset (4,477 construction cost samples)	Hybrid probabilistic ML(NGBoost-ETR), SHAP for explainability	Achieved high accuracy with uncertainty estimation, SHAP improved interpretability	Focuses on model performance, not early-stage prediction. No integration of macroeconomic indicators
8.	Das et al. (2024) - A comparative study of machine learning models for construction costs prediction with natural gradient boosting algorithm and SHAP analysis	Construction cost dataset (historical project data)	ANN, NGBoost, Linear Regression, SHAP for feature attribution and sensitivity analysis	Improved prediction accuracy and quantified uncertainty, explainable predictions	Limited focus on early-stage applications and lacks external economic factors

9.	Al Mnaseer et al. (2025)- Machine learning-aided time and cost overrun prediction in construction projects	Construction project data with cost/schedule performance indicators	ANN optimized with metaheuristics and ensemble learning (RF, XGBoost, LightGBM)	Addresses the early-stage prediction challenge and optimization techniques improve model performance	Lacks explainability
10.	Khodabakhshian et al. (2024)- Machine learning application in construction delay and cost overrun risks assessment	NYC School construction Authority dataset – 13570+ public school construction projects	Decision trees, XGBoost, ANNs and comparative performance evaluation	Comprehensive risk assessment framework using ML for both delays and cost overruns	Focuses on general risk, no explainability and no external factors
11.	Aung et al. (2023)- Using machine learning to predict cost overruns in construction projects	Construction dataset (limited size)	Baseline model - LR, Decision trees, RF, SVM	Demonstrates feasibility of ML for cost overrun prediction	Limited dataset, lacks explainability and no external economic factors integration

2.7 Research Gap

After reviewing the existing studies, it clearly shows that machine learning has improved the prediction of construction cost through different approaches like using advanced models, uncertainty quantification, and explainability techniques. These studies help to identify the important gaps in this research area.

First, early-stage models (Padala & Goyal, 2025; Deepa et al. (2025) show that it is possible to assess cost overrun risks during the early-stage of construction. Though these models improved their performance, they have not explored the integration of external economic indicators.

Second, there are models (Hamdan et al., 2025; Khodabakhshian et al., 2024; Al Mnaseer et al., 2025) which achieved high accuracy in prediction cost overruns which used execution phase data, and this limits the value for early-stage decision-making.

Third, advanced explainable machine learning models (Chen et al., 2026; Das et al., 2024; Gunarani et al., 2026) achieved accuracy in prediction and identified the key cost drivers that provide transparent, interpretable models because stakeholders require not only accuracy in predictions, but they would also like to understand which factors drive predictions. However, these studies did not focus on early-stage predictions and did not use external economic indicators.

Fourth, there are models (Darko et al., 2023; Alzara et al., 2025) which integrated the external economic factors into cost overrun prediction models and showed that this approach can enhance the model's performance, however there is lack of integration of these factors into early-stage project models.

Finally, all these studies show that there is a clear need for an integrated framework which combines early-stage data, external economic factors, and explainable machine learning techniques. This study will address these gaps by developing an integrated framework using publicly available construction project data to support informed decision-making before the construction begins, which helps to overcome cost overruns issue in the construction industry.

3. Research Questions

This research will answer the following key questions which addresses the gap in the existing studies.

1. Can machine learning models predict the cost overruns focusing on early-stage construction data and external economic indicators?
2. How do the external economic indicators such as inflation rates, construction market conditions influence and improve the prediction accuracy of the cost overrun prediction models?
3. Which machine learning model performs well by providing the highest accuracy for early-stage cost overrun prediction?
4. How can explainable AI techniques like SHAP will help to identify the key features which contribute to cost overruns that support decision-making?

4. Aim and Objectives

This research aims to develop an early-stage predictive framework for construction cost overrun predictions using the early-stage project data, and integrating them with the external economic indicators, then applying explainable AI techniques to improve interpretability and support decision-making.

The following are the objectives of this study,

- To develop multiple machine learning models for predicting cost overruns using early-stage data and comparing the models to analyze their performance.
- To integrate external economic indicators such as economic and market conditions and evaluate their impact on the model performance.
- To identify key drivers of cost overruns by applying explainability AI techniques (SHAP) to support transparency and decision-making for stakeholders.

5. Significance of the Study

One of the important challenges in the construction industry is cost overruns which leads to financial losses and delays. Improving the ability to predict cost overruns at an early stage helps for better planning and risk management. The significance of this research lies in addressing this issue by developing a cost overrun prediction model to enable early identification of cost overruns during early-stage using the pre-construction data. One of the key contributions of this research is integration of external economic indicators such as market and economic conditions, by incorporating these factors into the prediction model the study aims to improve the accuracy and reliability on the cost overrun prediction. Another major contribution is applying explainability AI techniques to identify the key features which influence the prediction. This helps to enhance the model interpretability and supports decision-making rather than just predictions.

The expected outcome is an early-stage prediction model that combines machine learning algorithms with external economic indicators to predict cost overrun classification using pre-construction data. This framework provides interpretable predictions using SHAP analysis which allows the project managers, project owners and schedulers to understand whether the project is at risk and explains the key factors that are driving to risk. By identifying the key cost drivers early, the stakeholders can decide on budget allocation, procurement strategies, and contingency planning.

This research has both national and international relevance, as the cost overruns are a challenging issue in construction industry worldwide. The early-stage prediction model, helps the organizations to plan on resource allocation, budget allocation and reduce failures, which gives confidence in project delivery. By using publicly available construction project data the proposed prediction model ensures that the approach is applicable to organizations across diverse construction environments and can be adapted to different regions by incorporating region-specific economic indicators. This research supports both academic and industry by addressing the unexplored areas in the industry. This research focuses on bridging the gap between prediction accuracy, external factors integration and interpretability in cost overrun prediction.

6. Scope of the Study

This study focuses on developing a machine learning model for predicting construction cost overruns at the early stage of a project. The scope of the research defines what the research will and will not address.

6.1. In Scope

The study will focus on the following elements:

- **Data source:** The study will use the publicly available construction project datasets, the BIM-AI Integrated dataset, Total Construction Spending Data Collection. These datasets provide project-level information such as project type, size, duration, location and initial cost estimates. The time-series economic data includes inflation rates, material price trends, and market activity indicators.
- **Feature Scope:** This study will include only pre-construction data and external economic indicators as inputs to the model
- **Approach:** This study will develop machine learning models using different algorithms like ensemble methods, gradient boosting techniques. Applying explainability AI techniques like SHAP to identify key drivers of cost overruns. The developed model's performance will be compared using several classification measures including Accuracy, Recall, Precision, F1-Score, AUC-ROC and Confusion matrix.
- **Geographic Scope:** This study will be using the publicly available construction project dataset which ensures that this approach can be applied to different environments and in the construction industry worldwide.

6.2. Out of Scope

This study excludes the following aspects from this study:

- **Execution-phase data:** This study will not use data like change orders, actual cost and progress tracking metrics which are available only during construction stage.
- **Real-time predictions:** This framework will not implement continuous model updating because real-time data integration is not performed and the cost estimation at the task or activity level is not used.
- **Qualitative risk factors:** The qualitative factors like organizational culture, relationships and political influences will not be modelled.
- **Deployment or Testing:** This study will not validate this framework on any active construction projects. No region- specific analysis will be included.

6.3 Reason for Defining the Scope

The scope is defined to make sure that this research will address a meaningful gap in the existing literature and focus on three primary reasons.

First, in the construction industry research area, there is lack of early-stage predictive models that integrate with external economic indicators and provide interpretability. By using only the pre-construction data, the research helps in decision-making before beginning the construction process.

Second, this study will not use the execution phase data which makes it clear that the model can be applied in real-world planning when such data is not available. Applying explainable techniques makes the framework more transparent and interpretable.

Third, using publicly available construction project datasets enhances the reproducibility and accessibility of the research and allows the other researchers to validate the results.

By clearly defining the scope of the research, it ensures that the research will be focused, achievable and aligned with its objectives in developing an early-stage cost overrun prediction model.

7. Research Methodology

This section describes the approach to achieve the research objectives. This study applies a methodology to develop an early-stage machine learning framework for predicting cost overruns. The methodology follows a workflow which consists of phases like data collection, data integration, data processing, feature engineering, model building, model evaluation and explainable analysis.

7.1 Research Design and Workflow

The research follows a seven-phase workflow to develop and validate an early-stage machine learning framework for construction cost overrun prediction classification.

The following Figure 1 shows the workflow of the research by transforming raw data to cost overrun prediction framework

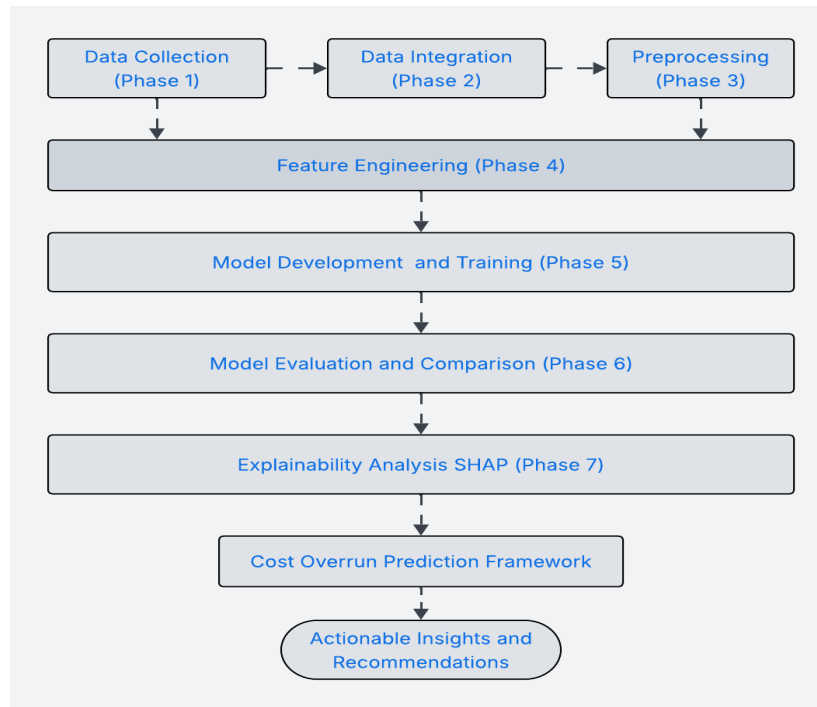


Figure 1: Simplified workflow of the research process (Source: Self)

7.2 Mapping Objectives and its Methods

This section helps to understand how each objective will be attained using a method with the references from the existing studies

Table 2: Objectives and Methods

Objective	Method	Reference
Objective 1: To develop multiple machine learning models for predicting cost overruns using early-stage data and comparing the models to analyse their performance	-Build machine learning models using different algorithms using only early-stage features like project, project type, initial budget, location and other characteristics -Then comparing and evaluating the model's performance using the appropriate classification metrics	-This approach of using early-stage data has been demonstrated by Padala & Goyal (2025) using ANNs on planning-phase data and Darko et al. (2023) using DNN and SVR on design-stage features
Objective 2: To integrate external economic indicators such as economic and market conditions and evaluate their impact on the model performance	-Conducting ablation study to build two model variants for each algorithm. One is the baseline model with only early-stage data, and the second variant is incorporating the external economic indicators along with the early-stage features -By calculating AUC-ROC, the performance difference will be measured and helps to understand how the economic indicators influenced the prediction	-Ablation study approach following standard machine learning evaluation has been demonstrated by Chen et al. (2026) -Integrating economic indicators has been applied by Alzara et al. (2025), who integrated macroeconomic indicators to construction cost forecasting.
Objective 3: To identify key drivers of cost overruns by applying explainability AI techniques (SHAP) to support transparency and decision-making for stakeholders	-Applying SHAP to the best performing model to identify the key cost drivers which influence the prediction. Evaluating the results and interpreting it to support decision-making	- Using SHAP explainability AI techniques has been demonstrated by several studies in construction cost research (Chen et al., 2025)

7.3 Data Collection

This research will use the publicly available construction datasets for developing the early-stage cost overrun prediction models.

The primary data source for this study is BIM-AI Integrated Dataset, which contains project-level information from different construction projects. The pre-construction features available in the dataset includes Project information, budget details, planned duration, awarded date and project characteristics. The data includes the historical information approximately between 2010 to 2024, the range ensures the balance to train the models and evaluate the model's performance.

The secondary dataset for this study is Total Construction Spending Data Collection and Related Sources, which contains external economic indicators and time-series data. This dataset includes Consumer Price Index (CPI), Producer Price Index (PPI), Total Construction Spending Index, Material-specific price indices, Labor wage indices. The indicators will be available in time-series format for the years 2010 to 2024 which matches the range of the project data.

7.4 Data Integration

The primary and the secondary datasets needs to be integrated before processing the data. For each project, the external economic indicators available during the early stage will be extracted through join operations from the secondary dataset. This approach is based on the methodology used by Alzara et al. (2025) for integrating macroeconomic data into construction cost.

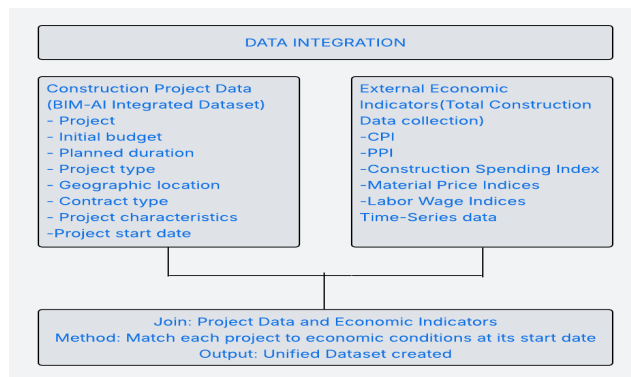


Figure 2: Illustrates the method of Integrating Datasets (Source: Self)

7.5 Data Preprocessing

Data preprocessing phase cleans and prepares the unified dataset for model training and makes sure it acts as a foundation for model development. The following steps will be applied, based on the established practices in construction cost prediction research (Padala & Goyal, 2025; Hamdan et al., 2025; Khodabakhshian et al., 2024). This phase includes the following steps:

- Handling missing values
- Handling Outliers
- Encoding the categorical features
- Feature Scaling

7.5.1 Target Variable Creation

We will have classification task performed in this approach,

- The cost overrun for each project will be calculated using the standard formula in construction cost management, (Hamdan et al., 2025; Darko et al., 2023; Padala & Goyal, 2025)

$$\text{Cost Overrun (\%)} = [(\text{Final Cost} - \text{Initial Budget}) / \text{Initial Budget}] \times 100$$

- This cost overrun% is used to create a binary classification target label,
Overrun = 1, if Final Cost > Initial Budget / Cost Overrun > 0%
No Overrun = 0, if Final Cost <= Initial Budget / Cost Overrun <= 0%

Classifying the predicted probabilities into risk categories will help to interpret the result and support decision-making. This approach is standard in construction cost overrun research (Hamdan et al., 2025; Gunarani et al., 2026). The three risk categories using threshold-classification are derived post-prediction as an interpretive tool for stakeholders.

Table 3: Risk Categories Interpretation

Predicted Probability	Risk Level	Recommended Action
<30%	Low Risk	Monitoring
30% - 60%	Medium Risk	Contingency adjustment
>60%	High Risk	Risk mitigation planning

7.6 Feature Engineering

This phase plays a very important role in the approach of developing a prediction framework. Derived features are created from the data which increases the predictive capability by capturing the patterns. This phase follows the feature engineering practices demonstrated in construction cost prediction research (Hamdan et al., 2025). Some other studies, (Das et al., 2024) also used feature engineering practices for prediction. This stage is one of the key contributions as project data is integrated with external economic indicators. After deriving the features, the feature importance analysis will be performed using Random Forest feature importance scores to identify the most important, redundant and most irrelevant features. Variance Inflation Factor (VIF) will be calculated to detect multicollinearity.

7.7 Model Development

The cost overrun classification prediction model will be developed using multiple machine learning algorithms and then compared using appropriate metrics to identify the best performance.

7.7.1 Algorithms to be evaluated

The following five supervised algorithms will be trained using the unified construction dataset followed by hyperparameter tuning to improve the performance of the model

- **Logistic Regression** will be used for building a baseline linear classification model to identify the interactions between the features. This algorithm calculates the probability of overrun using a logistic function, providing interpretable coefficients that indicate the direction and magnitude of feature effects on overrun likelihood (Aung et al., 2023)
- **Random Forest** is an ensemble method of constructing multiple decision trees to reduce overfitting. Using this method, we can build multiple decision trees to capture relationships between the project features and cost estimation through bootstrap aggregation (Gunarani et al., 2026)

- **XGBoost** is a powerful gradient boosting method which is good for strong performance on structured data. This approach sequentially builds decision trees where the errors in the previous level will be corrected by new trees, incorporating regularization techniques to prevent overfitting and achieve strong predictive performance (Hamdan et al., 2025)
- **LightGBM** is an efficient method for faster training and lower memory usage suited for high-dimensional feature spaces by optimizing computational efficiency through histogram-based learning and leaf-wise tree growth strategies (Al Mnaseer et al., 2025)
- **CatBoost** – powerful algorithm to handle categorical features through ordered boosting and novel encoding schemes that reduce the need for manual preprocessing of categorical variables such as project type, location, and contract method (Hamdan et al., 2025). CatBoost algorithm achieved the highest performance on construction project data with categorical features (Hamdan et al., 2025)

7.7.2 Training Strategy and Hyperparameter Tuning

The dataset will be split into training, test and validation subsets using stratified sampling for model development and validation.

Hyperparameter tuning will be conducted over key hyperparameters for all the algorithms to improve the model performance. Hyperparameters such as tree depth, learning rate and regularization strength will be tuned to prevent underfitting or overfitting to improve the model's performance (Das et al., 2024; Chen et al., 2026). K-fold cross-validation method will be applied by evaluating hyperparameter combinations by training and validating models on multiple data subsets (Hamdan et al., 2025). The best hyperparameter configuration will be selected using the metric, AUC-ROC score as it is good to measure classification performance across all probability thresholds (Khodabakhshian et al., 2024).

7.7.3 Ablation Study

Ablation study is a good practice in machine learning for isolating features contribution and similar approach used by Chen et al. (2026). For each algorithm two model variants will be developed, first model will be a baseline model which will be trained using only project-level features like budget, duration, project type, contract method and so on. The second model will be an enhanced model which will be developed using project-level features and external economic indicators like CPI, PPI, material price indices and derived features. Both models will be trained, and hyperparameter tuning will be done. Their performance difference will be calculated by calculating AUC-ROC for classification. This approach will show if integrating external economic indicators improve the prediction accuracy.

7.8 Model Evaluation

The developed models will be compared, and its performance will be assessed using the standard classification measures. The following measures will help us to understand the model quality and its predictive capability.

7.8.1 Primary Metric

AUC-ROC will be used as a primary metric for model ranking and comparison. This metric measures the model's ability to differentiate the projects that will face the cost overrun across the probability thresholds. The value ranges from 0.5 to 1.0 which will be suitable for early-stage cost overrun prediction because it is strong in class imbalance cases.

7.8.2 Supporting Metrics

- Accuracy: Proportion of Correctly classified projects out of total projects
- Precision: Proportion of projects that actually experienced overruns
- Recall: Proportion of actual overruns correctly identified by the model
- F1-Score: Calculates the mean of precision and recall giving a balanced measure
- Confusion Matrix: Detailed analysis of true positives, true negatives, false positives and false negatives

7.8.3 Validation Approach

The predicted probabilities will be mapped to risk categories for easier interpretation, and we can validate the outcome by calculating the actual overrun rate with each predicted risk category,

- What percentage of projects predicted as “High Risk” actually undergoes cost overruns?
- What percentage of projects predicted as “Low Risk” remains on budget?

7.8.4 Model Selection

All the models will be ranked based on AUC-ROC performance, the top ranked model will be considered for final evaluation and will be used to apply SHAP analysis. The model selection approach was implemented in Hamdan et al. (2025) research who compared fifteen algorithms.

7.9 Explainability Analysis Using SHAP

SHAP analysis improves interpretability by quantifying the contribution of different features that influences cost overruns. SHAP has been implemented in many studies, Chen et al. (2026) showed how the interpretability improves the model. Similarly, Gunarani et al. (2026) proved the importance of interpretability.

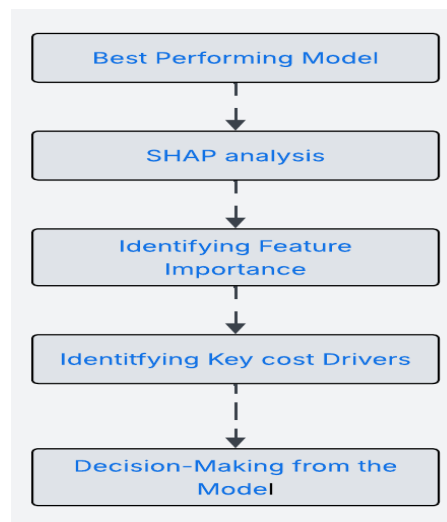


Figure 3: Explainable AI Workflow (Source: Self)

7.9.1 Global Interpretability

This method will identify which project features and external economic indicators acts as primary key drivers for the cost overrun predictions. Global SHAP analysis was demonstrated by Chen et al. (2026) and Das et al. (2024) in their studies. The features are ranked by SHAP values, and this will help us to compare the influence level of each feature. The SHAP values distribution will be visualized, and this shows the correlation between the features.

7.9.2 Local Interpretability

This method will identify insights for specific projects, which helps the project managers to understand why a specific project is at high risk. Local SHAP analysis was demonstrated by Gunarani et al. (2026) for sector-specific cost overrun analysis. The plots will decompose the prediction and identify the contribution of each feature, which will help to understand how the combination of features predict the cost overrun probability.

7.10 Expected Outcome

The methodology will develop an early-stage prediction of cost overrun classification framework which will predict cost overruns using pre-construction data integrated with external economic factors. The framework will map the predicted probabilities to risk categories which support decision-making and improve risk management in construction project planning. Risk categorization will help the stakeholders to identify the high-risk projects during the planning phase.

8. Requirement Resources

The following resources will be utilized to effectively implement this approach and evaluate the framework.

8.1 Data Sources

This study will use two publicly available datasets as primary sources which is available at no cost and no licensing requirements.

- [BIM-AI Integrated Dataset](#), consists of project level information
- [Total Construction Spending Data Collection](#), consists of external economic indicators

8.2 Software Resources

The software tools needed for this study are open-source and available at no cost, which avoids the licensing issues.

- Programming Language – Python
- Data Analysis: Pandas- data processing, Numpy- numerical operations
- Visualization: Matplotlib and Seaborn – SHAP plot generation and other visuals
- Machine Learning: Scikit-learn, XGBoost, LightBGM, CatBoost
- Explainability: SHAP library – interpretation and feature attribution analysis
- Environment: Google Colab, alternatively Jupyter Notebook
- Version Control: Git

8.3 Hardware Resources

The Hardware resources for this research provided below will sufficiently handle the datasets sizes and sufficient in training the machine learning algorithms.

- Computer or Laptop with at least 8GB RAM
- Multi-core Processor and 20 GB storage space
- Cloud-based alternatives such as Google Colab will be utilized at no cost

9. Research Plan

This research is planned for nine months from January to September 2026 covering 34 weeks of work schedule. This provides a structured timeline for completing all the major phases of the research from literature search through final submission. This approach ensures gradual progress enabling regular monitoring and assessment of the work. The following Table 5 shows the week-based research work which aligns with the program calendar.

Table 4: Research Workflow Plan (Week-based, January-September 2026)

Week	Month	Milestones	Tasks
1-2	January	Identifying domain and Research Area	Referring to the interested domain and identify the interested sub-domains
3-7	Jan-Feb	Literature review	Explore the existing studies on the domain selected and collect the research papers
8-12	March-April	Topic Confirmation	Select the topics on which the research is going to be performed along with the dataset and get the approval
13-20	April-May	Research Proposal Writing	Prepare a research proposal document explaining the objectives, methodology and outcome of the research
20-23	May-June	Data Collection and Integration	Collect the data and perform exploratory analysis to understand the data
23-25	June	Data Preprocessing	Clean and prepare the data to be ready for analysis
25-26	June	Feature engineering	Derive new features from the existing data and build a final feature set
24-27	June-July	Model development	Building five algorithms, training them and hyperparameter tuning will be done
25-28	June-July	Model Evaluation	Ablation study will be performed, and model evaluation will be done using the appropriate metrics
26-30	Jul-Aug	SHAP Analysis	Apply SHAP explainability AI techniques for interpretation
28-31	Jul-Aug	Results Analysis	The outcomes will be analysed to support decision-making
24-34	Jul-Sep	Dissertation writing	Dissertation will be written and submitted

The study has been organized into twelve phases to complete the research. The plan starts with identifying the interested domain and research area, identifying the gap in the existing studies and followed by topic confirmation and proposal, then implementing the methodology to develop the predicted classification framework. The development phase involves different activities like data collection, data preparation, model development and model evaluation followed by SHAP analysis. The results are then evaluated and measured to validate the effectiveness of the proposed solution. The below Figure 4 shows the Gantt Project Planner of the research activities, which highlights the planned and actual durations and the activities completion percentage. This chart shows the weekly distribution which helps to track effectively.

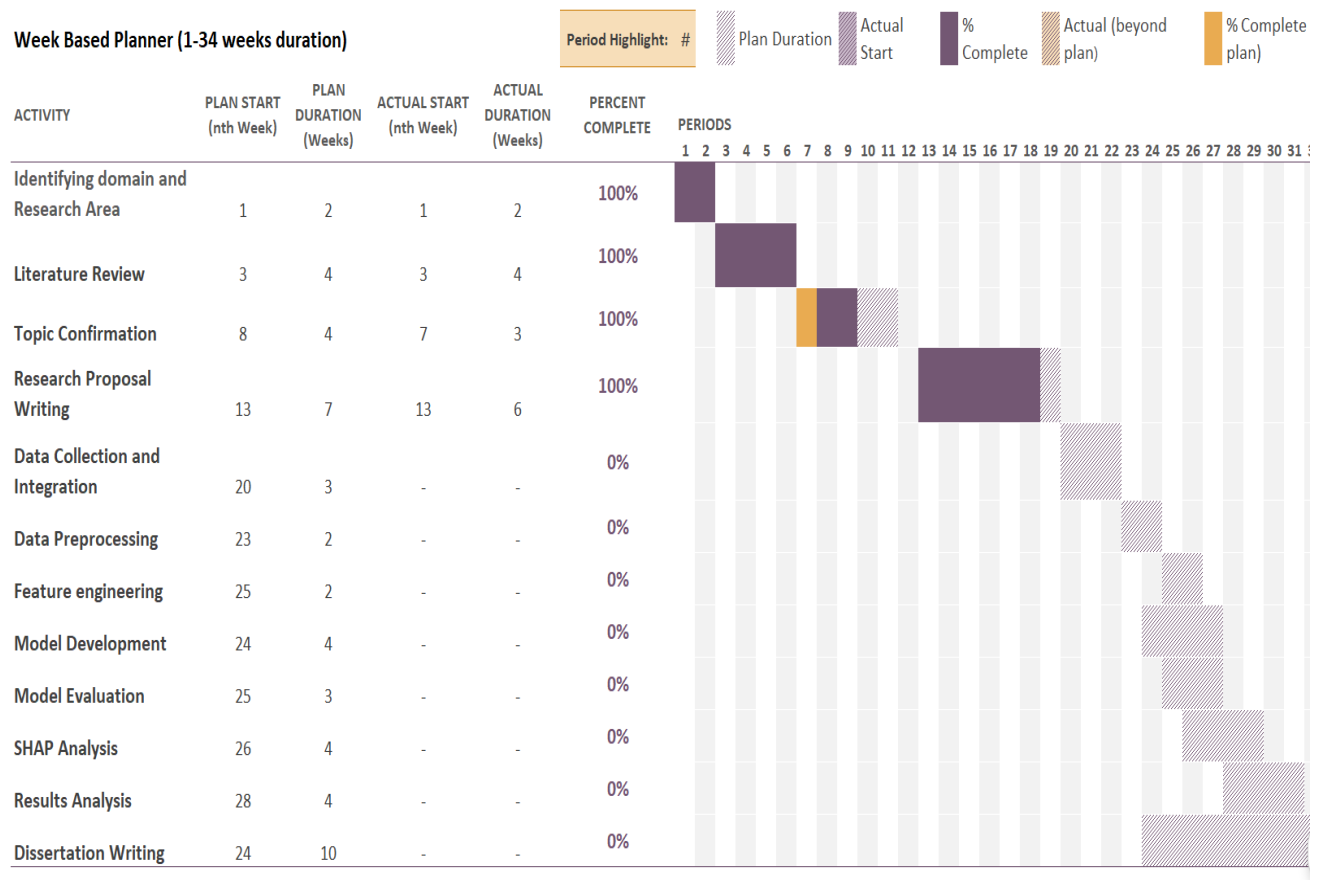


Figure 4: Research Planner -Week Based Gantt Chart

9.1 Risk Mitigation and Contingency Planning

After planning the schedule, the risks have been identified, and their mitigation strategies have been planned.

Model Performance: If the model performance is not to the expected level, then feature engineering will be reworked. For this iterative refinement, one week buffer between the Feature engineering and Model Development phase has been provided.

Time Constraints: The research plan has been planned to include overlapping phases wherever necessary to improve the results. The dissertation writing will be started in 24th week, to allow writing in parallel along with the research work to avoid rushed completion.

Supervisor Calls: Regular meetings will be scheduled with the supervisor to get their guidance and to identify any issues at the early stage of the work.

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